

# Baotong Tian

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## Education

### University of Rochester

*Ph.D. in Electrical and Computer Engineering*

GPA 4.0/4.0

Aug, 2024 – Aug 2029 (Expected)

### Tsinghua University

*Bachelor of Engineering in Automation*

GPA 3.86/4.0

Sep, 2020 – Jun, 2024

## Scholarships & Awards

- **2021, 2022 Tsinghua Arts and Culture Merit Scholarship** (Awarded to undergraduate students for great contribution to cultural activities within the department)
- **2022, 2023 Tsinghua Academic Excellence Scholarship** (Awarded to undergraduate students for excelling GPA throughout the school year)
- **2023 Tsinghua Innovation Award of Science and Technology** (Awarded to undergraduate students with excellent research potential)
- **2025 WIMIR Grant from ISMIR**

## Publications

- **Baotong Tian**, Cynthia Lu, Vincent Cheung, Ting-Kang Wang, Jonathan Churchill, Zhiyao Duan. "VIOLET: High-Fidelity Violin Synthesis with Techniques and Dynamics." Proc. ISMIR 2026.
- Zhang, Yu\*, **Baotong Tian\***, and Zhiyao Duan. "Conan: A Chunkwise Online Network for Zero-Shot Adaptive Voice Conversion." Proc. ASRU 2025. [[Arxiv](#)], [[Demo Page](#)], [[Code](#)]
- You, Zhang\*, **Baotong Tian\***, Lin Zhang, Zhiyao Duan. "PartialEdit: Identifying Partial Deepfakes in the Era of Neural Speech Editing." Proc. Interspeech 2025. [[Paper](#)], [[Project Page](#)], [[Dataset](#)]
- Borui, Zhang\*, **Baotong Tian\***, Wenzhao Zheng, Jie Zhou, and Jiwen Lu. "Exploring Unified Perspective For Fast Shapley Value Estimation." arXiv preprint arXiv:2311.01010 (2023). [[Arxiv](#)]

## Research Experience

### Expressive Violin Performance Rendering

Jun, 2025 - Present

*Research Assistant, Advisor: Zhiyao Duan*

- Created a paired dataset **CSV-TD** by automatically annotating expressive MIDI with note-level violin techniques and rendering corresponding audio with a commercial Kontakt sample library.
- Proposed **VIOLET**, a framework for generating violin performances with techniques and dynamics control.

### PartialEdit

Sep, 2024 - Jun, 2025

*Research Assistant, Advisor: Zhiyao Duan*

*Audio Information Research (AIR) Lab, Department of Electrical and Computer Engineering, University of Rochester*

- Created the partially deepfake dataset **PartialEdit** using advanced neural speech-editing models.
- Evaluated state-of-the-art utterance-level detectors and segment-level localization methods on this dataset, showing that models trained on traditional partially spoofed audio fail to detect speech edited by neural speech-editing models.
- Discussed how artifacts introduced by codec models affect the performance of models.

### Streaming Voice Conversion

Sep, 2024 - Present

*Research Assistant, Advisor: Zhiyao Duan*

*Audio Information Research (AIR) Lab, Department of Electrical and Computer Engineering, University of Rochester*

- Co-developed **Conan**, a chunk-wise online model for zero-shot voice conversion.
- Adapted to model timbre and style separately, and proposed a pixel-shuffle HifiGAN for streaming applications.

## Controllable Sheet Music Generation

June, 2021 - Dec, 2022

Research Assistant, Advisors: Xin Jin & Duo Xu & Song-chun Zhu  
AI Music Group, Beijing Institute for General Artificial Intelligence

- Learned music psychology and emotion perception in music.
- Learned about the generative theory and basic structural analysis of tonal music.
- Participated in the implementation of Music And-or-Graph.
- Developed a genetic algorithm for **music variation** in Python.

## Explainable Machine Learning

Feb, 2023 - Jun, 2024

Research Assistant, Advisors: Jiwen Lu & Jie Zhou  
Intelligent Vision Group, Department of Automation, Tsinghua University

- **Unified** current approaches of Shapley value estimation.
- Proposed **SimSHAP** as a simple and fast amortized Shapley value estimator.

## Work Experience

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### Streaming Text-to-Voice for Real-Time Preview and Voice Conversion

May, 2026 - Present

Research Intern, Mentor: Ke Chen, Yunyun Wang, Zeyu Jin  
Speech AI Team, Adobe Inc.

- Enhanced offline text-to-speech models with streaming capabilities to allow preview and generation in real time.

### Automatic Chord Recognition for Pop songs

Jun, 2023 - Aug, 2023

Algorithm Engineer Intern, Mentor: Jian Wu  
R&D Department, Beijing Deepmusic Technology Co.

- Applied Bi-LSTM for multi-task learning of bass and chord quality.
- Applied **Conditional Random Field (CRF)** as a post-filtering technique to enhance chord recognition task accuracy by incorporating context information.

## Teaching

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Teaching Assistant:

- ECE208/408 The Art of Machine Learning
- ECE270 Probability for Electrical Engineers

Spring 2025

Fall 2025

## Professional Services

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### Reviewer

Journal:

- IEEE Transactions on Audio, Speech, and Language Processing (TASLP) 2025
- IEEE Signal Processing Letters (SPL) 2026
- IEEE Transactions on Multimedia (TMM) 2026

Conference:

- IEEE Automatic Speech Recognition and Understanding Workshop 2025

### Volunteer

- ISMIR 2025 2025

## Music Background and Interests

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### Singing

- Joined the school choir at primary school.
- Won the 4th place in the Singing Carnival in High School.

### Instrument Playing

- Electric Piano (age 5 to 12, got the Amateur Highest Performance Level at the age of 12)
- Guitar (age 13 to present)

### Music Composition & Production

- Learned basic music production skills such as music arrangement and mixing in college.
- Selected for the Class of 2021-2022 in **Tsinghua University's Music Dream Program**, which is designed to nurture and develop campus musicians.
- Released 8 songs on QQ music & The NetEase's Cloud music, produced by **Logic Pro X**.

## Links

- Solo and Choir performance can be seen [\[here\]](#).
- My musician page on [\[The NetEase's Cloud Music\]](#), [\[QQ Music\]](#), and [\[SoundCloud\]](#).

## Programming & Language Skills

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### Programming Skills:

- Proficient: Python, PyTorch, Markdown, LaTeX, Git
- Familiar: MATLAB, C/C++, Linux, TensorFlow, HTML, etc

### Languages:

- TOEFL iBT: 104/120 (Reading 26, Listening 29, Speaking 24, Writing 25)
- GRE: 328/340 + 4.0/6.0 (Verbal 158, Quantitative 170, Analytical Writing 4.0)